

IBM Db2 Overview, Architecture & Use Cases DBMS Module 6

> **Definition:** IBM Db2 is an enterprise-grade Relational Database Management System (RDBMS) developed by IBM, engineered for high-performance transactional processing (OLTP), analytical workloads (OLAP), and hybrid data management (HTAP) across hybrid cloud environments.

1. Detailed Technical Explanation

1. Key Features of IBM Db2:

- **BLU Acceleration:** In-memory columnar processing for multi-fold speedup in analytical queries.
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2. IBM Db2 Product Editions

Db2 Edition	Target Environment	Hardware / Resource Limits	Key Features
Enterprise Edition	Large enterprises, mission-critical applications	Unlimited	Full functionality, including HTAP and advanced security
Express Edition	Small to medium businesses, development and testing	Up to 16GB memory, 10GB storage	Core DBMS functionality
Express-CE Edition	Cloud environments, open source	Unlimited	Open source, cloud-native architecture

3. Industry Use Cases

- Banking & Financial Services:** Core banking transactions, real-time credit card fraud detection using pureScale high availability.
 - Healthcare:** Patient data management, clinical research analytics.
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4. Comparison: IBM Db2 vs Other RDBMS

Feature	IBM Db2	Oracle Database	MySQL / PostgreSQL
Performance	High (BLU Acceleration)	High	Medium
Scalability	High (Scale-out)	Medium	High
Security	Advanced (Security Guard)	Advanced	Basic
Cloud Support	Yes (Cloud Pak for Data)	Yes (Oracle Cloud)	Yes

5. Core Concepts & Memory Keywords

- **pureScale:** IBM active-active database clustering for zero downtime.

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6. Must-Write Points for Exams

- IBM Db2 supports Hybrid Transactional and Analytical Processing (HTAP) in a single engine.

- Db2

7. Quick Recall Flow

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